

Structured Prompt Engineering Framework: Assessing The Effectiveness of The CRAFT Framework.

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Introduction

Prompt Engineering is the systematic design and refinement of inputs to large language models (LLMs) and other generative systems. It involves specifying goals, constraints, examples, and context in natural language, and sometimes, structured formats, to steer model behavior toward desired outputs [1]. A systematic survey of prompt engineering techniques highlights broad classification, including instruction prompting, in context learning, chain of thought prompting, role prompting, and tool augmented prompting, all of which can be composed into multi-step prompt chains and workflows [2].

In a 2023 field experiment conducted in collaboration with Boston Consulting Group (BCG) and researchers from Harvard Business School, MIT, and the Wharton School found that AI increased productivity and quality for tasks inside what they call the “jagged technological frontier,” but could hurt performance on tasks outside it [3]. The research found that workers given access to GPT-4 completed 12.2% more tasks and finished them 25.1% more quickly on average [3]. A parallel MIT study found that access to ChatGPT reduced task completion time by 40% and raised independently evaluated output quality by 18% [3].

While existing literature evaluates general structured input design [4] and automatic prompt optimization [5, 6], there remains a lack of empirical evaluation specifically focused on the C.R.A.F.T. (Context, Role, Actions, Format, and Tone [7]) assembly framework. This research seeks to evaluate if the specific five-parameter CRAFT framework provides a measurable performance increase in professional tasks. The central question this research addresses is: **To what extent does the CRAFT Framework improve output consistency and constraint adherence compared to unstructured prompting?**

Methodology

The methodology involves utilizing Automated Computational Benchmarking to assess the technical effectiveness of the CRAFT framework. A dataset of 50 standardized benchmark tasks, (e.g., HumanEval, DS-1000, FinBen) already pre-made and adapted from industry benchmarks, across different professions will be used

The procedure will involve assigning two prompt sets, the baseline which will be unstructured, natural language queries, and then using structured prompts generated by the CRAFT framework. Prompts will be tested across two distinct LLMs: Claude 3.5 Sonnet, GPT-4o.

Outputs will be scored using an advanced LLM either Gemini 1.5 Pro or Manus 1.6 Pro removing human bias.

Finally, the results will be compared across the two models to determine if CRAFT is model-agnostic or if performance gains vary significantly between architectures. The primary metric, y-axis, will be a Task Adherence & Accuracy Score (0-10), evaluated by a model based on a pre-defined rubric for each task.

Each output generated by test models will be fed into Manus 1.6 Pro or Gemini 1.5 Pro. The model will evaluate the response against a rubric based on the following three submetrics, each worth points toward the final 10-point score:

- (1) Constraint adherence (4 points): Did the model follow the specific format and tone instructions provided by the CRAFT framework?
- (2) Logical Accuracy awarded (4 points): Is the logic sound?
- (3) Completeness (2 points): Does the output address all elements of the Action and context parameters?

Significance of Findings and Future Work

This research addresses a gap at the intersection of human-computer interaction, organizational productivity, and AI literacy. While “The Prompt Report” identifies a vast taxonomy of over 58 distinct prompting techniques, it highlights the lack of “unified vocabulary” and standardized frameworks for non-expert users [2]. By validating the CRAFT framework, this study provides the bridge between complex academic taxonomies and practical professional application.

Reference list

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